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Studierichting: Informatica
Oefeningenreeks 5G nr. 11

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$$
\begin{aligned}
&y - r = \frac{R-r}{h}(x-0) \Leftrightarrow y = \frac{R-r}{h}x + r \\
&\Rightarrow \pi \int_0^h \left(\frac{R-r}{h}x + r\right)^2 dx = \pi \int_0^h \left(\frac{R^2 - 2Rr + r^2}{h^2}x^2 + 2r\frac{R-r}{h}x + r^2\right) dx \\
&= \pi \frac{R^2 - 2Rr + r^2}{3h^2}\left[x^3\right]_0^h + \pi r \frac{R-r}{h}\left[x^2\right]_0^h + \pi r^2 \left[x\right]_0^h \\
&= \pi h \left(\frac{R^2 - 2Rr + r^2 + 3Rr - 3r^2 + 3r^2}{3}\right) = \frac{1}{3}\pi h \left(R^2 + 2Rr + r^2\right)
\end{aligned}
$$

References